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FINAL

CENTRAL COALFIELDS LIMITED (CCL)

PROJECT REPORT OF KOTRE BASANTPUR PACHMO OCP (CAPACITY-5.0 MTY)

WEST BOKARO COALFIELD
DIST – RAMGARH & BOKARO (JHARKHAND)

[TEXT & APPENDICES]
(JOB NO. 3417145)

MARCH 2018

REGIONAL INSTITUTE-III
CENTRAL MINE PLANNING & DESIGN INSTITUTE LIMITED

(A Subsidiary of Coal India Limited)

GONDWANA PLACE, KANKE ROAD, RANCHI – 834 031,
JHARKHAND, INDIA

परिचालना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi

29/12/12



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PROJECT OFFICER
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सी.सी.एल. राँची
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2017/12

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[Signature]
20/12/18

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी.सी.एल., राँची
Ranchi, CCL, Ranchi

SUMMARISED DATA

Sl. No.	Particulars	Unit	Value
A.	GENERAL		
1	Name of Project		Kotre Basantpur Pachmo OCP
2	Name of Area / Company		Central Coalfield Ltd.
3	Nearest Railway Station from project	Name	Dania Railway Station
		km	4
4	Nearest National / State Highway / Approach road	Name	NH 33
		km	15

B.	GEOLOGICAL				
1	Name of geological blocks considered	Name	Kotre Basantpur	Pachmo	Total
2	Area of the geological blocks	sq. km	3.75	2.70	6.45
3	Borehole Density within blocks	BHs / sq.km	21.07	19.26	20.31
4	Description of all coal seams within block				

Stratigraphic Sequence	Thickness (m)	Net Geological Reserves (MT)	Remarks
XIII	1.29 - 4.54	1.201	
XII	1.10 - 2.86	1.752	
XI	2.48 - 5.20	4.669	
X	1.00 - 6.63	9.293	
IX	1.00 - 2.88	4.232	
VIII C	1.00 - 1.40	0.365	
VIII B	1.00 - 1.50	1.161	
VIII A	1.00 - 1.60	0.644	
VIII	1.30 - 3.45	9.378	
VI/VII	4.80 - 9.66	34.722	
VA	1.72 - 7.40	26.077	
V	1.19 - 9.65	36.780	
IV	1.00 - 7.40	34.035	
III	1.00 - 8.17	25.848	
IIA	1.00 - 4.80	6.615	
II Top	1.00 - 7.63	5.960	
II Bot	1.00 - 6.41	7.181	
II	1.00 - 12.45	22.023	
I	0.15-3.98	9.457	Not opencast able. About 44% reserve in Indicated category.
Local	0.10-3.78	8.998	Not opencast able. About 48% reserve in Indicated category.
'O'	0.40-3.85	-	Reserve not assessed in GR.
TOTAL		250.391	

Sl. No.	Particulars	Unit	Value
C.	TECHNICAL		
1	Area of the proposed mine block (only the quarry area)	sq. km	6.52
2	Borehole density within mine area	BHs/sq. km	20

Project Report for Kotre Basantpur Pachmo OCP (5.0MTY), CCL

3	Mine parameters Extent along strike (min. – max.) Extent along dip (min. - max.)	m	800 – 5500 500 – 1200
4	Description of coal seams proposed to be worked along with the parting details		

Name of seam/ Parting	Thickness range considered (m)	Av. Thickness / Parting Thickness (m)	Avg Ash %	Avg Grade	Mineable Reserve (MT)	Volume of OB (Mcum)
Top OB	4.65 - 70.02	27.36				80.77
Seam XIII	0.45 - 4.71	2.45	31.74	W IV	0.59	
Parting XIII - XII	2.31 - 31.2	10.23				4.42
Seam XII	0.25 - 3.94	1.97	30.32	W IV	0.80	
Parting XII - XI	28.79 - 67.05	43.67				17.25
Seam XI	2.48 - 6.38	3.96	34.44	W IV	2.91	
Parting XI - X	24.09 - 49.79	36.02				30.11
Seam X	0.45 - 6.63	4.31	29.74	W IV	5.92	
Parting X - IX	10.34 - 51.36	20.97				31.73
Seam IX	0.13 - 2.88	1.62	32.58	W IV	3.13	
Parting IX - VIII/VIII C	4.8 - 25.66	19.12				85.45
Seam VIII C	0.26 - 2.14	0.93	31.01	W IV	0.36	
Parting VIII C - VIII B	7 - 32.82	22.26				19.87
Seam VIII B	0.12 - 1.85	0.97	29.73	W IV	0.80	
Parting VIII B - VIII A	3.27 - 17.21	10.62				8.96
Seam VIII A	0.15 - 1.6	0.86	38.66	UG	0.63	
Parting VIII A - VIII	4.32 - 28.89	16.65				16.16
Seam VIII	0.59 - 4.65	2.26	30.61	W IV	6.44	
Parting VIII - VI/VII	32.11 - 63.68	50.05				137.90
Seam VI/VII	3.28 - 9.66	6.59	24.77	W III	27.17	
Parting VI/VII - VA	1.04 - 5.53	2.13				6.88
Seam VA	0.98 - 9.03	4.37	27.09	W III	17.46	
Parting VA - V	5.49 - 39.94	20.22				72.81
Seam V	0.66 - 9.65	5.20	32.07	W IV	22.92	
Parting V - IV	4.97 - 25.95	14.16				48.88
Seam IV	0.25 - 8.39	4.62	31.61	W IV	20.64	
Parting IV - III	2.4 - 39.44	22.48				83.62
Seam III	0.09 - 8.17	3.20	34.4	W IV	16.22	
Parting III - IIA	11.9 - 30.9	17.74				26.65
Seam IIA	0.1 - 4.8	2.30	38.85	UG	3.75	
Parting IIA - II Top	4.71 - 13.9	9.81				14.64
Seam II Top	0.71 - 3.33	2.01	36.36	UG	3.54	
Parting II Top - II Bot	0.62 - 7.66	2.75				2.52
Seam II Bot	0.97 - 5.2	2.64	40.32	UG	4.55	
Parting III - II	9.35 - 37.93	22.10				45.90
Seam II	0.1 - 12.45	4.85	39.72	UG	15.80	
Total	Coal OB				153.63	734.53

Sl. No.	Particulars	Unit	Value
5	Av. Stripping Ratio	m ³ /te	4.78
6	Method of Mining		By Shovel Dumper Combination

Project Report for Kotre Basantpur Pachmo OCP (5.0MTY), CCL

Sl. No.	Particulars	Unit	Value
7	Target Output	MTY	5.00
	Production capacity (at 125%)	MTY	6.25
	Production capacity (at 85%)	MTY	4.25
8	Year of achieving Target Production	Year	7 th
9	Year of start of Internal Dumping	Year	6 th

10	Production Phasing (date upto target year)			MT			
Year	1	2	3	4	5	6	7
Coal (MT)	Construction period			0.40	1.50	3.00	5.00
OB (Mcum)				1.31	5.73	10.91	18.45

11	Variants proposed	Variant-I	Coal production & OB removal to be done departmentally.
		Variant-II	Coal production & OB removal to be done by outsourcing.

12	Total Mine Life	Years	36	
	Pre-construction period	Years	0	
	Construction period	Years	3	
	Production build-up period	Years	3	
	Production period	Years	29	
	Tapering / mine closure period	Years	1	
13	Major HEMM Deployed (Max)	Unit	Variant-I	Variant-II
	Diesel Hydraulic Shovel	10-12 Cum	7	-
	Diesel Hydraulic Shovel	5.5-6.5 Cum	7	-
	Diesel Hydraulic Backhoe	4.3-5.0 Cum	1	-
	Rear Dumper	100 T	72	-
	Rear Dumper	60 T	82	-
	Diesel RBH Drill	250 mm	5	-
	Diesel RBH Drill	160 mm	8	-
	Dozer	410 HP	6	-
	Dozer with Ripper Attachment	410 HP	3	-
	Dozer	700-880 HP	7	
14	Total Manpower	Nos	1326	163
15	Overall Output per manshift (OMS) – (on max manpower)	Tonnes	14.28	-

16	Seam-wise weighted average grade of coal (non-coking/coking)	Non-Coking /Coking	Grade
	Seam XIII	Coking	W IV
	Seam XII	Coking	W IV
	Seam XI	Coking	W IV
	Seam X	Coking	W IV
	Seam IX	Coking	W IV
	Seam VIII C	Coking	W IV
	Seam VIII B	Coking	W IV
	Seam VIII A	Coking	Ungraded
	Seam VIII	Coking	W IV

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
महोदय, रांची
WIV, C.C.L. Ranchi

Project Report for Kotre Basantpur Pachmo OCP (5.0MTY), CCL

16	Seam-wise weighted average grade of coal (non-coking/coking)	Non-Coking /Coking	Grade
	Seam VI/VII	Coking	W III
	Seam VA	Coking	W III
	Seam V	Coking	W IV
	Seam IV	Coking	W IV
	Seam III	Coking	W IV
	Seam II A	Coking	Ungraded
	Seam II Top	Coking	Ungraded
	Seam II Bot	Coking	Ungraded
	Seam II	Coking	Ungraded

Sl. No.	Particulars	Unit	Value
17	Presence of Major Surface Constraints (nallas, road, power line, etc.)	Nalas & Nadi	Chutua nala in the South and Baghraiya nala in south-east.
18	Coal Transport within the mine (In-pit belt conveying system or by Truck)		Dumper, In-pit belt conveyor and HAC
19	Surface Coal Transport to Siding/Despatch Point and Mode of Despatch		By conveyor to nearby Washery (existing/ upcoming).
20	Any Railway Siding and distance		NA
21	Name of any Specific Customer/Industry		Basket linkage

D. ENVIRONMENTAL & OTHERS

Sl. No.	Particulars	Unit	Variant-I	Variant-II
1	Civil Construction Residential houses Housing satisfaction	Nos. %	729 55	90 55
2	Water Demand Colony Industrial	kl	670.63 1158.11	88.27 786.56
3	Total Land required Land acquired Total Land to be acquired Non-Forest land Forest land	Ha	1220.39 0.00 1220.39 405.46 814.93	1209.39 0.00 1209.39 394.46 814.93
4	Land to be acquired within minetake area (excavation area) Non-Forest land Forest land	Ha	651.61 271.52 380.09	651.61 271.52 380.09
5	Land to be acquired outside minetake area (Beyond Excavation Area, such as Approach Road, Infrastructure, Colony, etc.) Non-Forest land Forest land	Ha	188.38 115.92 72.46	177.38 104.92 72.46
6	Land to be acquired for external dumping Non-Forest land Forest land	Ha	380.39 18.02 362.38	380.39 18.02 362.38
7	Net Present Value of Forest Land Total Area Total Value	Rs.Lakhs/Ha Ha Rs.Lakhs	18.32 814.93 14929.48	

Project Report for Kotre Basantpur Pachmo OCP (5.0MTY), CCL

Sl. No.	Particulars	Unit	Variant-I	Variant-II
8	Habitation & Rehabilitation No. of villages within mine boundary No. of PAFs to be rehabilitated	Nos.	8 1000	
9	Cost of land & Rehabilitation Total Cost R&R only	Rs. crores	304.03 109.27	298.74 109.27
10	Total EMP Capital Social cost Environment cost	Rs. crores	523.22 270.90 252.32	489.71 260.43 229.29
11	Make of Water (Maximum)	cum/day	295440	
12	Total installed pumping capacity	lps	900	
13	Drainage of the Area (Name of river/nala)		Chutua nala, Baghraiya nala draining out to Bokaro River	

E.	FINANCE			
Sl. No.	Particulars	Unit	Variant-I	Variant-II
1	Total Capital Investment		2146.56	855.88
	Upto Target Year	Rs. crores	1514.88	744.61
	Beyond Target Year		631.68	111.27
2	Total Capital Investment on P&M		1116.47	26.15
	Upto Target	Rs. crores	609.01	16.35
	Beyond Target		507.46	9.80
3	a) Capital for Sanction (Target Year)		1514.88	744.61
	b) Capital for Sanction (Project Life)	Rs. Crores	2146.56	855.88
4	Earnings per manshift (EMS) (target year)	Rs.	3230.78	3663.02
5	Estimated Cost of Production			
	At 100% production level	Rs. / tonne	1744.14	1531.87
	At 85% production level		1925.03	1584.65
6	Estimated average selling price	Rs. / tonne	2301.00	2301.00
7	Estimated Profit			
	at 100% production level	Rs. / tonne	556.86	769.13
	at 85% production level	Rs. / tonne	375.97	716.35
8	Financial Internal rate of return (FIRR)			
	At 100% production level	%	26.05%	40.18%
	At 85% production level		18.02%	34.96%
9	Break-even point			
	Production	Mty	3.24	1.39
	Production level	%	64.76%	27.73%
10	Cost of Outsourcing (average)			
	OB	Rs/cum	-	117.70
	Coal	Rs/tonne	-	75.49
11	Final Mine Closure Cost / te.	Rs/tonne	11.89	12.16
12	Year of Achieving target Production	Year	7th	7th

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PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
CCL Ranchi

List of Appendices

S N	Appendix No.	Title of the Appendix	Variant-I	Variant-II
1	A	Statement showing Estimated total capital investment and its phasing	1X	1X
2	A.1	Estimated capital investment on land acquisition	2X	2X
3	A.2	Estimated capital investment on buildings	3X	3X
4	A.2.1	Estimated capital investment on residential buildings	4X-5X	4X
5	A.2.2	Estimated capital investment on service buildings	6X-7X	5X
6	A.2.3	Building cost index	8X	Ref Var-I
7	A.3	Estimated capital investment on plant and machinery with phasing	9X	6X
8	A.3.1	Estimated capital investment on P&M - HEMM	10X	-
9	A.3.2	Estimated capital investment on P&M - Electrical	11X-13X	7X-10X
10	A.3.3	Estimated capital investment on P&M - Workshop and stores	14X-28X	-
11	A.3.4	Estimated capital investment on P&M - pumps, pipes and fittings	29X	Ref Var-I
12	A.3.5	Estimated capital investment on Coal Handling Plant	30X-34X	Ref Var-I
13	A.3.6	Estimated capital investment on other P&M	35X	Ref Var-I
14	A.3.7	Estimated capital investment on Telecommunication	36X	11X
15	A.4	Estimated capital investment on Furniture & Fittings	37X	12X
16	A.5	Estimated capital investment on Railway Siding	38X	Ref Var-I
17	A.6	Estimated capital investment on Vehicles	39X	13X
18	A.7	Estimated capital investment on Prospecting & Boring	40X	Ref Var-I
19	A.8.1	Estimated capital outlay in mines	41X	14X
20	A.8.2	Estimated capital investment on roads and culverts	42X-47X	15X-20X
21	A.8.3	Estimated capital investment on water supply and sewerage arrangements	48X-53X	21X-26X
22	A.9	Statement showing revenue expenditure capitalized	54X	27X
23	B	Statement showing Job-wise / Category-wise requirement of manpower	55X-60X	28X-31X
24	B.1	Statement showing estimated salaries and benefits	61X	32X
25	C	Estimated avg cost of production per te & profitability	62X	33X
26	C.1	Estimated cost of production at 100% capacity utilization	63X	34X
27	C.2	Estimated cost of production at 85% capacity utilization	64X	35X
28	C.3	Statement Showing Economics at Different Capacity Utilisation	65X	36X
29	C.4	Statement showing Sensitivity of IRR	66X	37X
30	D-D.1	Cash flow statement for financial analysis at 100% and 85% of the capacity.	67X-68X	38X-39X
31	E	Estimated capital investment on Reclamation	69X	-
32	F	Environmental Protection Measures	70X-71X	40X-41X

List of Plates

Sl No	Particulars	Drawing no.	Plate No	Scale
GEOLOGY				
1	Geological Map of West Bokaro Coalfield	RI-3/OCM/001431	1	NTS
2	Geological Plan of Basantpur-Kotre and Pachmo Blocks	RI-3/OCM/001432	2	1:5000
3	Topographical Plan of Basantpur-Kotre and Pachmo Blocks	RI-3/OCM/001433	3	1:6000
4	Floor Contour Plan of Seam-II/II Bottom	RI-3/OCM/001434	4	1:5000
MINING				
5	Final Stage Quarry Plan	RI-3/OCM/001435	5	1:6000
6	Cross-section A-A', B-B', & C-C'	RI-3/OCM/001436	6	1:2000
7	Cross-section D-D', E-E' & F-F'	RI-3/OCM/001437	7	1:2000
8	Final Stage Dump Plan	RI-3/OCM/001438	8	1:6000
9	Box Cut Plan (1st Year)	RI-3/OCM/001439	9	1:6000
10	3 rd Year Stage Plan	RI-3/OCM/001440	10	1:6000
11	5 th Year Stage Plan	RI-3/OCM/001441	11	1:6000
12	13 th Year Stage Plan	RI-3/OCM/001442	12	1:6000
13	Land Use Plan	RI-3/OCM/001443	13	1:10000
14	Post Mining Land Use Plan	RI-3/OCM/001476	14	1:10000
ELECTRICAL & MECHANICAL				
15	Layout of Unit Workshop & Store	R3/E&M/M03052	15	NTS
16	Single Line Diagram of Substation 33/6.6kv, 2x10MVA	R3/E&M/E03225	16	NTS
17	Key Plan of CHP	R3/E&M/M03067	17	1:10000

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सी.सी.एल., राँची
C.C.L., Ranchi

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PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi


20/12/12

परियोजना अधिकारी
PROJECT OFFICER

कोटा - गतपुर पंचायत कोटा माईन
Kotla - Ghatpur Panchayat Kotala Mine
सी.सी.एल., रांची
C.C.L., Ranchi

CHAPTER-I INTRODUCTION

1.1 BACKGROUND OF THE PROJECT

Kotre-Basantpur and Pachmo coal mine has been assigned to Central Coalfields Limited (CCL)/ Coal India Limited (CIL) in the year 2017 by MoC. It comprises two geological blocks of the West Bokaro Coalfield (WBCF) namely Kotre-Basantpur block and Pachmo block. The findings of the exploration in Kotre-Basantpur and Pachmo blocks indicated existence of coking coal.

The Kotre-Basantpur geological block, having an area of 3.75 sq.km, is located in the northern part of the WBCF and is a part of the Hazaribagh area of CCL falling in Ramgarh district of Jharkhand. The Kotre-Basantpur block, named after the local villages is a potential property lying immediately north of the present Kedla Project of CCL.

The Pachmo geological block, having an area of 2.70 sq.km, is located in the northern part of WBCF and falls in Bokaro district of Jharkhand. The Pachmo block, named after the local village is a potential property lying immediately north of the present Loiyo & Jharkhand Projects of CCL.

A Mining Plan of Kotre-Basantpur and Pachmo coal block was prepared by CMPDI in May 2007 for Tata Steel. However as the scope of mining plan has changed for CCL, primarily due to change in coal evacuation route and introduction of "High Angle Conveyor (HAC)" for coal transportation, it necessitated preparation of a fresh Project Report. Although, the present PR takes input from the earlier Mining Plan.

The "Project Report of Kotre-Basantpur Pachmo OCP, CCL (5.0MTY)" has been prepared based on the "Geological Report on Basantpur-Kotre Block" prepared by CMPDI in Feb 1997, "Geological Report on Pachmo Block" prepared by CMPDI in Sept 1998. This report envisages mining of seams up to Seam-II through Opencast method.

1.2 EXPLORATION STATUS

A brief summary of the agency-wise year of drilling, number of boreholes, meterage and borehole density is given below:


परियोजना अधिकारी
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सी०सी०एल०, राँची
C.C.L., Ranchi

Exploration Agency	Borehole Series	Period of Operation		Drilling	
		From	To	No. of boreholes	Meterage
Kotre-Basantpur block (within the block)					
NCDC	NCWBK	Jan-67	May-68	2	694.79
GSI	WBKB	Apr-80	Jun-80	1	365.15
CMPDI	CMK	Apr-78	Aug-79	1	198.77
CMPDI	CMB	Mar-87	Mar-90	53	8659.15
CMPDI	CMB	Nov-92	May-93	19	3053.35
Total				76	12971.21
Kotre-Basantpur block (outside the block)					
NCDC	NCWBK	Mar-65	Aug-65	2	699.01
GSI	WBKB	Apr-80	Jun-80	1	368.35
CMPDI	CMK	May-78	Aug-78	4	702.15
CMPDI	CMB	Mar-87	Mar-90	14	3045.75
CMPDI	CMB	Nov-92	May-93	1	321
CMPDI	CMPM	May-93	Jan-94	2	621.8
Total				24	5758.06
Grand Total for Kotre-Basantpur block				100	18729.27
Pachmo block (within the block)					
GSI	WBKB	Jan-79	Jun-79	2	466.15
CMPDI	CML	Apr-76		1	176.75
CMPDI	CMB	Mar-87	May-93	6	1433.15
CMPDI	CMPM	Mar-90	May-95	41	8536.8
Total				50	10612.85
Pachmo block (outside the block)					
IBM	WBL	Nov-63		1	417.2
NCDC	NCWBK	Mar-64	Dec-66	2	996.95
CMPDI	CML	1976	1978	7	1157.46
GSI	WBKB	Jan-79	Jun-79	2	733.5
CMPDI	CMK	Sep-84	Feb-86	4	1236.05
CMPDI	CMB	Mar-87	May-93	3	788.9
CMPDI	CMPM	Mar-90	May-95	3	382.05
Total				22	5712.11
Grand Total for Pachmo block				72	16324.96

The borehole density of the Kotre-Basantpur and Pachmo Geological Blocks are as follows:

Block	Area (sq km)	No of Boreholes	BH Density
Kotre-Basantpur	3.75	79	21.07
Pachmo	2.70	52	19.26
Overall	6.45	131	20.31

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी. सी. लोहा रांची
R-3, CMPDI

1.3 PRESENT STATUS OF THE MINE/PROJECT:

Coal mining activity has not yet started in these blocks. Mining activity in the adjoining blocks Kedla and Jharkhand in the south is undergoing. The block is covered by different land types viz., forest, tenancy, GMK, GMA and JJ.

1.4 JUSTIFICATION OF THE PROJECT:

The liberalisation of coal sector by Govt. of India has generated wide spread interests for Private and Public sector investments in power generation and steel production. This has resulted in a sharp increase in the demand of coking and non-coking coal in CCL. Opening of this Kotre Basantpur Pachmo (KBP) OCP (5.0MTY) is therefore, proposed with a view to fulfill the above indicated growth in demand. Beside this Project has to be developed as per the time limit given by MoC in their Allotment Agreement.

1.5 SALIENT FEATURES OF PRESENT PR (CAPACITY, CAPITAL, ETC.)

The normative and peak production of the project is 5.0MTY of Washery-IV grade coal. The present report envisages two Variants. The capital is given in Appendix-A of respective Variants. The broad details of the various variants proposed in the PR are as follows:

Variant-I

In Variant-I, coal production and OB removal has been envisaged to be done departmentally. Crushed and sized coal (-100mm) will be dispatched to nearby washery. Crushing of coal till 12 years will be done at Pit top and later coal will be brought to surface by High Angle Conveyor (HACs) after in-pit crushing. The variant proposes departmental coal crushing and conveying.

Variant-II

In Variant-II, coal production, OB removal has been envisaged to be outsourced. The coal crushing & conveying (including application of HACs), pumping, workshop and security are also proposed to be outsourced.

During the initial years, the ROM coal from the OCP will be sent to Kedla Washery situated at a distance of about 600m from the proposed storage bunker. In later years, the excess coal will be sent to upcoming Tapin Washery.

As suggested by CCL during the discussion of Draft PR, for evacuation of washed coal from Kedla Washery, provision for 10.25 km long railway siding from Kedla Washery to Dania Railway station (in the East Central Zone of Indian Railways) has been made. (Please refer Appendix-A.5).

1.6 DIFFICULTIES AND CONSTRAINTS IN MINING WITH ASSOCIATED RISK

- i) Kotre Basantpur Pachmo OCP is a virgin mine. The block is covered by forest and non-forest land. The area north of the block is marked by metamorphic, whereas, that in the west by Barren Measures. The topography of the area is more or less gently undulating and a number of seasonal nalas (Kotre, Jhumra, Pachmo, Baghraiya along with other streams) emerge from the high range (Hazaribagh plateau) which flow towards south through the mining area and meets west to east flowing Chutua nala and Baghlata nala. Flow of water from the nalas, into the mining area needs to be arrested by constructing and maintaining garland drains along the quarry periphery. In addition to these, four nallas (Kotre, Jhumra, Pachmo & Baghraiya) needs to be diverted away from the mining area as and when the mine approaches them. Financial provision for a total length of 11.7 km nala diversion has been made in the Project Report. The scheme for nala diversion prepared for TATA STEEL is under revision at CMPDI.
- ii) The projectised area is structurally complicated having a total of 11 major faults (throw varying from 0-300m). The gradient of the strata is quite steep. It varies from 8deg to 25deg. However in major part it is 8deg to 15deg.
- iii) A huge quantity of OB will be generated during mining. With steep gradient of the strata, concurrent internal dumping will be restricted. A large area for external OB dump will be required. The OB has been proposed to be dumped in the metamorphic, to the north of Kotre Basantpur block. The topography of the area available for OB dump is hilly (RL gradually increases from 360m in the northern quarry boundary to 410m further north) thereby restricting OB accommodation.
- iv) Steep gradient of the quarry restricts internal OB dumping quantities and thus increases external OB dump requirement. To overcome this difficulty it is proposed that the benches are formed along dip and concurrent backfilling may be started as soon as a portion of the mine reaches its dip side boundary limit. However when benches are formed along rise to dip, partial outsourcing may not be possible as the same bench will encounter both coal and OB and it will not be possible to demarcate area for either coal outsourcing or OB outsourcing.
- v) Embankment along the Chutua nala for a stretch of about 2 km is proposed as a safety measure.

1.7 PROJECT OBJECTIVES AND TARGET BENEFICIARIES

The primary objective of the mine is to meet the gap between demand and supply of washery grade coal as projected from the West Bokaro Coalfield and CCL as a whole. The secondary objective is the socio-economic development of the villagers residing in and around the Kotre Basantpur Pachmo OCP.

The targeted beneficiaries of the project include all stakeholders but not limited to the following:

- a) Central Coalfields Limited, Ranchi
- b) Coal India Limited, Kolkata
- c) Steel Plants linked with the mine,
- d) All the future ancillary industries coming up in the area
- e) People residing in the villages within or near the project boundary viz: Basantpur, Kotre, Pachanda, Hurdag, Rahawan, Pachmo, Baghraiya, Ichakdih and Purnapani villages.
- f) The State Govt. of Jharkhand
- g) Heavy Earth Moving Equipment manufacturing companies in India


परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi

 20/12/12

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतणर पंचमो कोल माईन
Kotre Basant, Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi

CHAPTER-II

MARKETABILITY & JUSTIFICATION

2.1 DEMAND AND SUPPLY SCENARIO

The availability and demand from CCL is given below. (Please refer Table on "Demand and Supply of Coal")

2.2 UTILITY OR MARKET FOR THE COAL FROM PROJECT

Liberalisation of power, steel & other sector by Govt. of India has generated wide spread interests for Private and Public sector investments in these sectors. As such, there is an appreciable increase in the number of upcoming industries in both Private and Public Sectors. This has resulted in a sharp increase in the demand of coal (coking and non-coking both) in CCL. Project Report of Kotre Basantpur Pachmo OCP (5.0MTY) is therefore, proposed with a view to partially fulfil the above indicated growth in demand.

2.3 AVAILABLE LINKAGE OR FIRM FUEL SUPPLY AGREEMENT (FSA)

The coking coal from the Kotre Basantpur Pachmo OCP is proposed to be linked to nearby Kedla Washery.

2.4 JUSTIFICATION OF OPENING THE PROJECT BASED ON MARKETABILITY (TO FULFILL THE GAP IN DEMAND) AND/OR FIRM LINKAGE (FSA)

The availability figures till end of 2019-20, as shown in the following Table, have been obtained from CCL. It appears from the figures that till 2019-20 the ACQ/ Commitment of CCL is well above the targeted production of CCL. The production from the upcoming project will help in meeting ACQ/ Commitment of CCL and justifies implementation of this project.


परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
G.C.L., Ranchi
RI-3, CMPDI

Demand and Supply of Coal

Fig in Million tonnes

YEARS	2017-18		2018-19		2019-20	
POWER HOUSE COMMITMENT	ACQ	Commitment	ACQ	Commitment	ACQ	Commitment

PRE NCDP FSA	31.6	28.44	31.6	28.44	31.6	28.44
LOA ROUTE FSA	21.51	16.13	21.51	16.13	21.51	16.13
BRIDGE LINKAGES	5.89	5.89	5.89	5.89	0	0
IMPORT SUBSTITUTION (NTECL Vallur and TANGEDCO) LOA route	1.25	0.94	1.25	0.94	1.25	0.94
EXPECTED ACQ FOR 2016-17 DUE TO ACHIEVEMENT OF COD (Barh Unit 1 and BRBCL)	2	1.5	0	0	0	0
TOTAL POWER HOUSE	62.25	52.9	60.25	51.4	54.36	45.51

CPP	1.14	0.69	0.11	0.06	0.00	0.00
CPP(LOA)	0.86	0.46	0.79	0.39	0.34	0.17
SPONGE IRON	0.37	0.22	0.04	0.03	0.00	0.00
SPONGE IRON (LOA)	0.76	0.38	0.7	0.35	0.09	0.04
STEEL	1.18	0.71	0.12	0.07	0.00	0.00
OTHERS (including Fertiliser AND State Agency))	2.3	1.38	1.08	0.65	0.89	0.53
AUCTION LINKAGE	0.82	0.61	0.82	0.61	0.82	0.61
TOTAL OTHER THAN POWER HOUSES	7.43	4.45	3.66	2.16	2.14	1.35

TOTAL ACQ AT CCL	69.68	57.35	63.91	53.56	56.5	46.86
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EXPECTED COD (Power House)						
BARH STPP	3.33	2.5	10	7.5	10	7.5
BHARTIYA RAIL BIJLEE	2.5	1.88	5	3.75	7.5	5.63
NTPC KANTI	1.6	1.2	1.6	1.2	1.6	1.2
NAVINAGAR (BSEB)	7	5.25	7	5.25	7	5.25
BOKARO A	0	0	0	0	1.98	1.49
NORTH KARANPURA	0	0	0	0	4	3
TOTAL	14.43	10.83	23.6	17.7	32.08	24.07
TOTAL ACQ AT CCL	84.11	68.18	87.51	71.26	88.58	70.93


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परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi

Project Report for Kotre Basantpur Pachmo OCP (5.0MTY), CCL

YEARS	2017-18		2018-19		2019-20	
	ACQ	Commitment	ACQ	Commitment	ACQ	Commitment
SAIL (RAW COKING COAL)						
SAIL (WASHED COAL)						
RINL (WASHED COAL)						
TOTAL MOU						
TOTAL FSA + MOU AT CCL	84.11	68.18	87.51	71.26	88.58	70.93

PRODUCTION TARGET AT CCL	80	104	127
COKING COAL PRODUCTION TARGET	22.55	23.39	25.51
COKING COAL DEMAND	16.19	13.40	-

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi, CMPDI

CHAPTER-III

PROJECT SITE INFORMATION

3.1 LOCATION

3.1.1 Broad location of project area in relation to geological block and coalfield with latitudes and longitudes

Kotre-Basantpur block and Pachmo block is located in the northern part of West Bokaro Coalfield. **Kotre-Basantpur** block, having an area of 3.75 sq km, is under the command area of Hazaribagh Area of Central Coalfields Limited in Ramgarh District, Jharkhand. The block is bounded by latitudes 23°48'30" and 23°49'24" North and longitudes 85°34'18" and 85°36'05" East. **Pachmo** block, having an area of 2.70 sq km, lies in Bokaro District of Jharkhand and is bounded by latitudes 23°48'06" and 23°48'58" North and longitudes 85°36'28" and 85°37'57" East. Both the blocks are covered in Survey of India toposheet no. 73E/9 (in 1:50000 scale).

3.1.2 Area of the Project

The projectised block covers an area of 11.51 sq. km., which includes area for external OB dump and infrastructure along with the quarry. The maximum quarry length in east-west direction is 6.0 km whereas the same in north-south direction is 1.2 km.

3.1.3 Limiting boundaries of the quarry area

East: Eastern block boundary of Pachmo Block.

West: Fault F2 and incrop of Seam-II.

North: Fault F1 forms the floor boundary.

South: Chutua nala / proposed railway siding Kedla Washery to Dania railway station along the Chutua nala.

3.2 ACCESSIBILITY AND COMMUNICATION

Kotre-Basantpur block is connected to National Highway No. 33 between Ranchi and Hazaribagh by an all-weather metalled road leading from Charhi to Kedla washery. This road terminates in the south-western corner of the block near Kedla washery. The block is also connected by few Kutcha seasonal roads. However, an all-weather road connection to the interior of the block does not exist.

Pachmo block lies adjacent to Kotre-Basantpur block in the east. Their common boundary falls in Pachmo nala. The block is approachable by a 2 km long dry weather Kutcha road from Loiyo-Charhi metalled road near Ichakdih village. This road crosses Chutua Nala immediately east of Ichakdih village. The block is also connected by a few Kutcha seasonal roads. The office of the General Manager, Hazaribagh Area of CCL at Charhi, is located at a distance of about 15km from the south west corner of Kotre-Basantpur block.

The Dania Railway station is about 4 kms from the eastern boundary of the block. It is located near the foothill of the Lugu hill in the North- Eastern part of the West Bokaro Coalfields on Gomoh Barkakana loop line of the East Central Zone of Indian Railways.

The block is about 40km from Hazaribagh & about 95km from Ranchi via Charhi. The nearest commercial airport is situated at Ranchi.

3.3 CLIMATE AND RAINFALL

The climate is tropical with severe summer. The temperature during summer (March to June) goes as high as 45°C. The summer days are hot with dusty winds but nights are generally pleasant. The minimum summer temperature is around 20°C. The winters (November to February) are cold and the minimum temperature recorded is 4°C.

The rainy season is generally from June to October. The total rainfall in a year on an average is about 1200 mm of which about 80% of the precipitation is during rainy season.

3.4 TOPOGRAPHY WITH DRAINAGE PATTERN OF AREA

The Kotre-Basantpur and Pachmo blocks lie between Hazaribagh plateau in the north and Damodar valley in the south. The Lugu hill (peak 976m), is only 10km to the east. The topography of the area is more or less flat with gentle undulations which at times is dissected by ravines and gullies at places. There are numerous seasonal nalas emerging from the high range (Hazaribagh plateau) which flow mostly towards south and forms prominent nalas and meet the major nalas of the block.

The area to the north of blocks are marked by metamorphics. The ground elevation varies from 305m (in the south) to 365m (in the north). The general elevation varies from 330 to 350m in the area and the general slope of the ground is towards south.

Chutua nala, a prominent tributary to Bokaro River, flows roughly west to east near the southern boundary. North to south flowing Pachmo nala, Kotre nala and Jhumra nala joins Chutua nala in the area. To the east of Pachmo nala, another prominent nala, Baghraiya nala flows from north to south. It meets Baghlata nala in the south east of the area which finally joins

Bokaro River. The Chutua nala is more or less perennial in nature for a major part of the year except for hot summer days. Rest nalas of the block become dry during summer season but they experience flash floods during rainy season. Drainage of the block is mainly controlled by Chutua nala and to some extent by Baghlata nala. Both of these nala ultimately discharges the total run-off to Bokaro River in the east.

3.5 PRESENT LAND USE PATTERN

The projectised area is covered by different land types viz., forest, tenancy, GMK, GMA and Jungle Jhari and cultivated land. The area covered by the forestland is shown in Land Use Plan of the project. The total forest area within the projectised area (including external OB dump) comes to 7.91 sq. km.

3.6 HUMAN SETTLEMENT

There are eight (8) village within the proposed leasehold boundary of the Project. No. of households with population within each of the village as per the Census-2011 is given below:

Name of Village	Population	Household
Basantpur	2243	424
Kotre	0	0
Pachanda	742	122
Pachmo	1928	456
Hurdag	1124	260
Rahawan	1363	292
Baghraiya	72	16
Purnapani	222	51

None of the villages fall completely within the Project leasehold boundary. The Project affected Families (PAF) and Project affected person (PAP) have been roughly estimated as 1000 and 4600 nos. respectively.


परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी.सी.एल. राँची
C.C.L., Ranchi

पिकेडोप कलकत्ता
मिशनरी सोसायटी
कोतरे बसंतपुर पंचमो कोल माईन
सी.सी.एल., राँची


20/11/12

परियोजना अधिकारी
PROJECT OFFICER
कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी.सी.एल., राँची
C.C.L., Ranchi

CHAPTER – IV

GEOLOGY AND DEPOSIT APPRAISAL

4.1 BACKGROUND / INTRODUCTION:

4.1.1 Geological Reports Prepared:

The regional exploration of Basantpur-Kotre and Pachmo blocks were taken up by GSI during 1977. A total of eleven boreholes were drilled and the geological mapping of area was also done on 1:8000 RF. Following reports by GSI are available.

1. Interim report on regional exploration for coal in Basantpur-Kotre, Pachmo and adjoining area, West Bokaro Coalfield, Distt. Hazaribagh, Bihar (1979).
2. Second interim report on regional exploration for coal in Basantpur-Kotre, Pachmo and adjoining area, West Bokaro Coalfield, Bihar (1983).

The Pachmo block was taken up for detailed exploration by CMPDI in March 90 and 41 boreholes were drilled within the block and three boreholes drilled outside the block in Jharkhand block.

Detailed exploration by drilling in Basantpur-Kotre block by CMPDI was taken in June 86. The block is approachable only during dry season. The drilling done in different phases is as under:

1st phase – March 87 to July 87.

2nd phase – Dec. 87 to June 88

3rd phase – Jan. 89 to July 89

4th phase – Jan. 90 to Mar. 90

In Basantpur-Kotre block, a total of 46 boreholes have been drilled by CMPDI involving a total meterage of 7804.25 meters. Earlier NCDC had drilled 2 boreholes involving a total meterage of 694.79 meters. GSI has also drilled one borehole in the block with a total meterage of 365.15 meters.

The shallowest and deepest boreholes in Basantpur-Kotre block are CMB-20 and CBM-67 with 29.70m and 422.40m respectively. A total of 13 boreholes were drilled per sq km including the boreholes drilled by GSI and NCDC. The average depth of the boreholes is 180.90 meters.

4.1.2 Block Boundaries:

The Basantpur-Kotre and Pachmo Block combined having an area of 6.45 sq. km. is located in the northern part of the coalfield and falls in the Ramgarh Dist. and Bokaro district of Jharkhand respectively. Both the blocks are virgin from mining point of view and lies between Lat. $23^{\circ}48'06''$ to $23^{\circ}48'58''$ and Long. $85^{\circ}34'18''$ to $85^{\circ}37'57''$ E.

The block is covered by Survey of India toposheet No.73 E/9(1:50000) and topo sheet no. 3

on 4"=1 mile. The area is covered by aerial photographs on RF 1:10000.

The Pachmo block is contiguous to Loiyo and Jharkhand block in the east and Basantpur-Kotre block in the west. The area north of the combined block is marked by the metamorphic and fault F1. The boundary between Basantpur-Kotre and Pachmo is along the southerly flowing Pachmo Nala which is one of the most important tributaries of the Chutua Nala.

Limits Of Basantpur-Kotre and Pachmo Block Combined:

Table: 4.1 Geographic Co-Ordinates of Combined Blocks boundaries

Extensions	Latitude	Longitude
Minimum	23°48'06" 30"	85° 34 '18" ✓
Maximum	23°48'58"	85°37'57" ✓

Table: 4.2 Geological Markers showing combined blocks boundaries

Limits	Description
Northern	Marked by metamorphics.
Western	Marked by the trace of fault F2 on the floor of Local Seam occurring below Seam-I.
Southern	Marked by U/G trace of fault F3 (in floor of seam-I) having almost E-W to NW-SE trend.
Eastern	An arbitrary line joining CMPM-43 along the incrop of Seam I and terminating at the northern block boundary.

4.1.3 Other Details:

The investigation by GSI in Basantpur-Kotre and Pachmo block have also indicated the existence of very strong coking coal in the area.

The Basantpur-Kotre block has been divided into seven sectors mostly on the basis of faults and surface features. Pachmo Block has been divided into three sectors.

Table: 4.3 Description of Sectors, Basantpur –Kotre Block

Sector	Limiting Boundary	Description
Sector-A	North	Barakar Metamorphic contact
	South	Fault F1
	East	Pachmo Nala
	West	Fault F2
Sector-B	North	Fault F1
	South	Fault F2
	East	Fault F2
	West	Fault F2
Sector-C	North	Fault F2
	South	Fault F3
	East	Pachmo Nala
	West	Metamorphic Contact

परियोजना अधिकारी
PROJECT OFFICER

कोतरे बसंतपुर पंचमो कोल माईन
Kotre Basantpur Pachmo Coal Mine
सी०सी०एल०, राँची
C.C.L., Ranchi

Sector	Limiting Boundary	Description
Sector-D	North	Fault F3
	South	Fault F5
	East	Pachmo Nala
	West	Fault F8
Sector-E	North	Fault F5
	South	Chutua Nala
	East	Pachmo Nala
	West	Fault F8
Sector-F	North	Fault F6
	South	Kedla Fault/Chutua Nala
	East	Fault F6
	West	Fault F8/ Fault F7
Sector-G	North	Metamorphic contact
	South	Fault F7
	East	Fault F8
	West	Fault F7

Table: 4.4 Description of Sectors, Pachmo Block

Sector	Limiting Boundary	Description
Sector-A	North	Fault F7
	South	Fault F3
	East	Fault F3
	West	Fault F3
Sector-B	North	Fault F2 & F4
	South	Fault F7
	East	Fault F1/ block boundary
	West	Block boundary
Sector-C		Fault F2 & F1/ northern block boundary

4.2 EXPLORATION STATUS:

4.2.1 Number of boreholes:

In total 172 boreholes (100 in Kotre Basantpur & 72 in Pachmo Block) have been drilled in different phases from 1963 to 1995. Total meterage of 35054.23m have been drilled so far by various agencies. The details are given in Table 4.5.

Table: 4.5 No of Boreholes and Meterage Drilled by Various Agencies

Exploration Agency	Borehole Series	Period of Operation		Drilling		
		From	To	No. of boreholes	Meterage	B.H. details
Kotre-Basantpur block (within the block)						
NCDC	NCWBK	Jan-67	May-68	2	694.79	
GSI	WBKB	Apr-80	Jun-80	1	365.15	
CMPDI	CMK	Apr-78	Aug-79	1	198.77	
CMPDI	CMB	Mar-87	Mar-90	53	8659.15	
CMPDI	CMB	Nov-92	May-93	19	3053.35	
Total				76	12971.21	
Kotre-Basantpur block (outside the block)						
NCDC	NCWBK	Mar-65	Aug-65	2	699.01	
GSI	WBKB	Apr-80	Jun-80	1	368.35	
CMPDI	CMK	May-78	Aug-78	4	702.15	
CMPDI	CMB	Mar-87	Mar-90	14	3045.75	
CMPDI	CMB	Nov-92	May-93	1	321	
CMPDI	CMPM	May-93	Jan-94	2	621.8	
Total				24	5758.06	
Grand Total for Kotre-Basantpur block				100	18729.27	
Pachmo block (within the block)						
GSI	WBKB	Jan-79	Jun-79	2	466.15	WBKB-3, 6
CMPDI	CML	Apr-76		1	176.75	CML-40
CMPDI	CMB	Mar-87	May-93	6	1433.15	CMB-43, 45, 48, 71, 72, 73
CMPDI	CMPM	Mar-90	May-95	41	8536.8	CMPM 1 to 11, 13, 15 to 41, 44, 93
Total				50	10612.85	
Pachmo block (outside the block)						
IBM	WBL	Nov-63		1	417.2	WBL-24
NCDC	NCWBK	Mar-64	Dec-66	2	996.95	NCWBK-17, 37
CMPDI	CML	1976	1978	7	1157.46	CML-1, 3, 32, 34, 35, 36, 37
GSI	WBKB	Jan-79	Jun-79	2	733.5	
CMPDI	CMK	Sep-84	Feb-86	4	1236.05	CMK-167, 183, 201, 203
CMPDI	CMB	Mar-87	May-93	3	788.9	
CMPDI	CMPM	Mar-90	May-95	3	382.05	CMPM-12, 14, 42
Total				22	5712.11	
Grand Total for Pachmo block				72	16324.96	
Grand Total				172	35054.23	

4.2.2 Density of boreholes: The borehole density of the Kotre-Basantpur and Pachmo Geological Blocks are as follows:

Block	Area (sq km)	No of Boreholes	BH Density
Kotre-Basantpur	3.75	79	21.07
Pachmo	2.70	52	19.26
Overall	6.45	131	20.31

Table: 4.6 Density of Boreholes for Different Seams

Seam Name	Boreholes Intersection	Combined Area (sq km)	Borehole density (bh per sq km)
XIII	6	6.45	0.93
XII	9	6.45	1.40
XIA	11	6.45	1.71
XI	15	6.45	2.33
XA	14	6.45	2.17
X	22	6.45	3.41
IXA	17	6.45	2.64
IX	27	6.45	4.19
VIII C	32	6.45	4.96
VIII B	38	6.45	5.89
VIII A	43	6.45	6.67
VIII	47	6.45	7.29
VII C	43	6.45	6.67
VII B	27	6.45	4.19
VII A	32	6.45	4.96
VII/VI COMB.	55	6.45	8.53
VII/VI/A COMB.	10	6.45	0.16
VA	56	6.45	8.68
V	69	6.45	10.70
IV	73	6.45	11.32
IIIA	9	6.45	1.40
III	70	6.45	1.09
IIA	24	6.45	3.72
II	35	6.45	5.43
II TOP	18	6.45	2.79
II BOT	16	6.45	2.48
II(T+B)	7	6.45	1.09
I	44	6.45	6.82
LOCAL	28	6.45	4.34
SEAM 'O'	14	6.45	2.17

Note: VI/VII has merged with Seam VA in two isolated area in the central part of Kotre-Basantpur as encountered in borehole nos. CMB-37 & 59.

4.2.3 Number of Boreholes Analyzed:

The ultimate analysis of the seams in Kotre-Basantpur are available from few boreholes only. Detail washability test for major seams including detail caking properties on cleans and petrographic studies have been given. In Pachmo Block, washability tests were carried out on the coal cores of BH No. CMPM-20 and 21 for seams IIA, III, IV and V. Petrographic analysis was carried out on raw coal basis for the coal seams encountered in BH Nos. CMB-67 and 73 and CMPM-4, 5, 7, 8, 9, 23, 26, 30, 36. Petrography on washed coals of seam IV and V was also carried out for borehole Nos CMPM-20 and 21.

The details of the analysis conducted for the blocks are furnished in following tables.

Table No: 4.7 Details of band by band analysis as carried out by different agencies on coal cores of boreholes drilled in Basantpur-Kotre block, West Bokaro Coalfield.

S N	Analytical agency	Details of Boreholes	Total no. of boreholes analysed
1.	Coal Survey Laboratory, Namkom, Ranchi	CMB-24,35,36,38,41,45,47,49,52,54,56,57,59,60-69,75-98	47
2.	Chemical Laboratory, CMPDIL, Ranchi	CMB-22,23,25,27,28,30,31,33,37,39,40,42,44,50,51,70,74	17
3.	M/s Coal Inspection Service, Ranchi	CMB-13,15,18,19,100	5

Table No: 4.8 Details of band by band analysis as carried out by different agencies on coal cores of boreholes drilled in Pachmo block, West Bokaro Coalfield.

S N	Analytical Agency	Details of boreholes	Total no. of Boreholes analysed
1.	Coal Survey Laboratory, Namkum	WBKB-3,6 CML-40 CMB-43,45,48,67 CMPM-1-9, 20-22, 37-41,43	25
2.	Chemical Laboratory, CMPDIL, HQ	CMB-71,72,73 CMPM-10,11,13,15,23-27,29-31,33,35,36	18
3.	SUPCO	CMPM-16,17,18,19	4

4.3 GEOLOGY AND STRUCTURE OF BLOCK:

4.3.1 Brief Geological Setting:

The Basantpur-Kotre and Pachmo blocks are situated on the northern limb of the prominent northern syncline of the West Bokaro Coalfield. The axial region of the synclinal structure roughly passes in East-West direction. This axial region extends to Pachmo block in the South

Western part of the block. The reversal of dips have been observed on the basis of borehole data. This data includes Borehole nos. CMB-43 & 71, CMPM-23, 5 & 16 in Pachmo block. In Kotre-Basantpur Block these boreholes are located in 3 pockets in the southern part of the block. These pockets are around WBKB-9, CMB-39, 43 & 71 CMB-40, 65 & 100 and CMB-60, 97 & 98.

Table: 4.9 Stratigraphic sequence of Basantpur-Kotre and Pachmo Block Combined

Formation	Lithological Units	Drilled thickness Range
Alluvium	Sandy Soil	22.60-24.00 meters
Barren Measures(Found Mainly in Kotre-Basantpur Block)	Grey shale, Carb/Sandy Shales with thin coal bands& Iron stone shales	110m
Barakar Formation	Medium to coarse grained sandstone pebbly in the lower portion with shale sandy shale & coal seams	413.00-430.00m
Kharharbari Formation	Mainly arkosic sandstone pebbly sandstone, thin coal band one coal seam.	75-161m
-----Unconformity-----		
Pre-Cambrian/Metamorphics	Granite Gneiss, Chlorite, mica schist, quartzite, pegmatites etc	Encountered at depth range of 9.45m to 176.95m

4.3.2 Coal bearing formations and their general behavior:

The Gondwana column available in the Pachmo and Kotre-Basantpur block is represented by rocks of Barakar, Barren Measure and Karharbari Formation. Coal seams are occurring in Barakar and Karharbari formations. Barren Measures are absent in Pachmo block unlike Basantpur-Kotre Block while Barakar, Karharbari are found in both the blocks viz Pachmo and Basantpur-Kotre. The Barakars are well developed in this coalfield and contains the potential coal seams designated as I to XIII. Beside this there are a number of thin coaly horizons which attain workable thickness in certain parts and have been named as A, B, C of the main seams like VIIA, VIIB, VIIC, VIIIA, VIIIB, VIIIC, etc. The Karharbari contains only one coal seam called 'O' seam. The sequence of coal seams and their intervening parting are established from GSI, NCDC and CMPDI boreholes.

The surface mapping as well as the borehole records of the Basantpur-Kotre block does not indicate the existence of any intrusive in the block either in the form of dyke or sill.

4.3.3 Sequence of Coal seams and partings with in the block:

The youngest coal seam XIII, due to structural disposition has been encountered only in few boreholes. Among the younger seams, Seam X and XI are comparatively thick with thin

intervening parting. The seam incrop has a strike length of about 1 km in sector D and is potential for opencast mining in the block. Below seam X a number of thin seams having a thickness of about 1m occur. Seam VIII is most potential from thickness and quality point of view. Seam VA, IV and III are potential seams of West Bokaro Coalfield and are attractive from quarriable point of view. Due to structural disposition however, in Basantpur Kotre the opencast potentiality of these seams have been considerably reduced. Seam III and I are inferior in quality and erratic in thickness.

The only seam available in Karharbari formation i.e. seam O has a thickness of around 3-4 m in Basantpur-Kotre block. Though the quality of the seam is better, its potentiality in Basantpur-Kotre block is restricted due to presence of numerous strike and oblique faults.

The sequence of coal seams and partings as established in the block are given in the following table:

Table: 4.10 Sequence of coal seams and partings in Kotre-Basantpur & Pachmo blocks

Seam Name	Thickness Range (m.)		Generalised Range (m)	Remarks
	Min. (Bh No.)	Max. (Bh No.)		
XIII	1.29	4.54	2-4	Faulted-3Bhs
	(CMB-40)	(CMB-31)		
Parting	4.15	31.20	5-15	
	(CMB-31)	(CMB-98)		
XII	0.82	2.86	2-2.5	Faulted-3bhs
	(CMB-31)	(CMB-98)		
Parting	2.60	21.94	5-15	
	(CMB-97)	(CMB-60)		
XIA	0.15	2.10	0.15-0.90	Faulted-3bhs, Not dev.-3bhs
	(CMB-43)	(CMB-97)		
Parting	2.80	34.84	6-25	
	(CMB-23)	(CMB-97)		
XI	2.48	4.52	3-4	Faulted-2bhs
	(CMB-100)	(CMB-71)		
Parting	2.80	37.16	6-25	
	(CMB-23)	(CMB-39)		
XA	0.30	1.28	0.4-1.0	Faulted-3bhs, Not dev.-2bhs
	(WBKB-9)	(CMB-40)		
Parting	12.05	28.78	20-27	
	(CMB-57)	(NCWBK-37)		
X	0.45	6.63	3-6	Faulted-3bhs
	(CMB-57)	(CMPM-16)		
Parting	2.40	14.48	4-10	

Seam Name	Thickness Range (m.)		Generalised Range (m)	Remarks
	Min. (Bh No.)	Max. (Bh No.)		
	(CMPM-23)	(CMB-57)		
IXA	0.10	1.17	0.5-1.0	Faulted-3bhs, Not dev.-6bhs, Carbshale-1bh
	(CMPM-22)	(CMB-60)		
Parting	2.93	23.65	8-15	
	(CMB-57)	(CMPM-10)		
IX	0.13	2.88	1-2.50	Faulted-1 bh, Not dev.-1 bh
	(CMB-97)	(CMB-35)		
Parting	4.80	25.66	15-24.0	
	(CMPM-44)	(CMPM-1)		
VIIIC	0.26	2.14	0.7-1.20	Faulted-1hb, Not dev. 3bhs
	(CMPM-07)	(CMB-24)		
Parting	9.85	32.82	15-30	
	(CMB-64)	(CMPM-23)		
VIIIB	0.12	1.85	0.7-1.50	Faulted-1bh, Not dev-2bhs
	(CMB-37)	(WBKB-03)		
Parting	3.27	17.21	10-12	
	(CMB-57)	(CMB-64)		
VIIIA	0.15	3.50	0.5-1.20	Incrop-1bh, Not dev-1bh
	(CMPM-20)	(CMB-97)		
Parting	4.32	28.89	10-25.0	
	(CMB-24)	(CMPM-07)		
VIII	0.59	4.65	1.20-3.0	
	(CMPM-22)	(CMPM-23)		
Parting	2.20	31.47	8-15	
	(NCWBK-37)	(CMK-119)		
VIIC	0.13	1.91	0.50-1.00	Not. Dev-7bhs, Carb Shale-2bhs
	(CMB-39)	(CMPM-18)		
Parting	3.38	37.82	4.0-12.00	
	(CMB-50)	(CMB-24)		
VIIIB	0.15	3.33	0.30-1.25	Not.Dev-21 Bhs, Incrop-1bh, Carbshale-5bhs
	(CMB-74)	(CMK-119)		
Parting	9.35	32.49	7.0-20.0	
	(CMB-96)	(CMPM-20)		
VIIA	0.20	2.00	0.5-1.0	Faulted-1bh, Not dev.-30bhs, Carbshale-1bhs(WBKB-3)
	(CMPM-08)	(CMB-41)		

Seam Name	Thickness Range (m.)		Generalised Range (m)	Remarks
	Min. (Bh No.)	Max. (Bh No.)		
Parting	0.58	6.65	1.50-3.50	
	(CMPM-04)	(CMPM-10)		
VII/VI COMB.	3.05	9.66	5.0-7.0	Faulted-1bh
	(CMK-119)	(CMB-42)		
VII/VI/VA COMB.	5.74	13.37	8.00-12.00	
	(CMPM-15)	(CMPM-23)		
Parting	1.04	4.38	1.5-2.50	
	(CMPM-06)	(CMB-57)		
VA	1.12	7.40	4.0-6.0	Faulted-1bh, Floor faulted-1bh, Incrop-3bhs
	(CMB-50)	(CMPM-10)		
Parting	5.49	46.51	9-35.0	
	(CMB-54)	(CMK-119)		
V	2.19	9.65	4-6.50	Faulted-3bhs, Weathered-1bh, Roof faulted-4bhs
	(CMB-13)	(CMB-42)		
Parting	4.97	24.91	8-20.0	
	(CMB-75)	(CMB-63)		
IV	0.25	8.39	4.0-6.0	Faulted-5bhs, Incrop-2bhs
	(CMB-37)	(CML-40)		
Parting	10.20	22.12	12-20	
	(CMPM-21)	(CMPM-08)		
IIIA	0.15	1.40	0.30-0.40	Not dev-16 bhs, Faulted-1bh
	(CMPM-20)	(CMPM-33)		
Parting	5.26	11.88	6.0-7.0	
	(CMPM-08)	(CMPM-21)		
III	0.09	8.17	2.0-5.0	Incrop-1bh, Not dev-1bh
	(CMB-30)	(CMB-62)		
Parting	11.90	30.90	15-20	
	(CMPM-30)	(CMPM-36)		
IIA	0.10	4.80	2-4	Not Dev-2bhs, Carbshale-1bh
	(CMPM-39)	(CMPM-20)		
II	0.26	12.45	5-7	Floor faulted-2bhs, Incrop-1bh Carbshale-1bh
	(CMB-96)	(CMB-70)		
Parting	4.45	8.75	3-8	